Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
2	*(a)*			If object A exerts a force on object B, then B exerts an equal and opposite force on A (1) Accept action and reaction are equal and opposite The wall applies a force on the car in the collision (1) Therefore the car applies an equal and opposite force on the wall (1)	1	1 1		3		
	(b)	(i)		Use of $a = \frac{v-u}{t}$ and $F = ma$ (1) $a = \frac{(0-15)}{0.028}$ = (–)535.7 [m s^{-2}] (1) [ignore sign] Accept 536 [m s^{-2}] F = 85 × 535.7 **ecf** = 46 000 [N] (1)	1	1 1		3	2	
		(ii)		Gradient / slope of the graph isn't constant / acceleration is changing or increasing (1) Don't accept acceleration is decreasing Which shows that the force increases through the stopping process (1) Don't accept that the force changes		2		2	2	
	(c)			The crumple zone increases the distance or time to stop (1) So the force is smaller [on the driver] (1) N.B. Award only 1 mark if a fully correct answer is accompanied with an incorrect statement about work done changing	1 1			2	2	
				Question 2 total	**4**	**6**	**0**	**10**	**6**	**0**